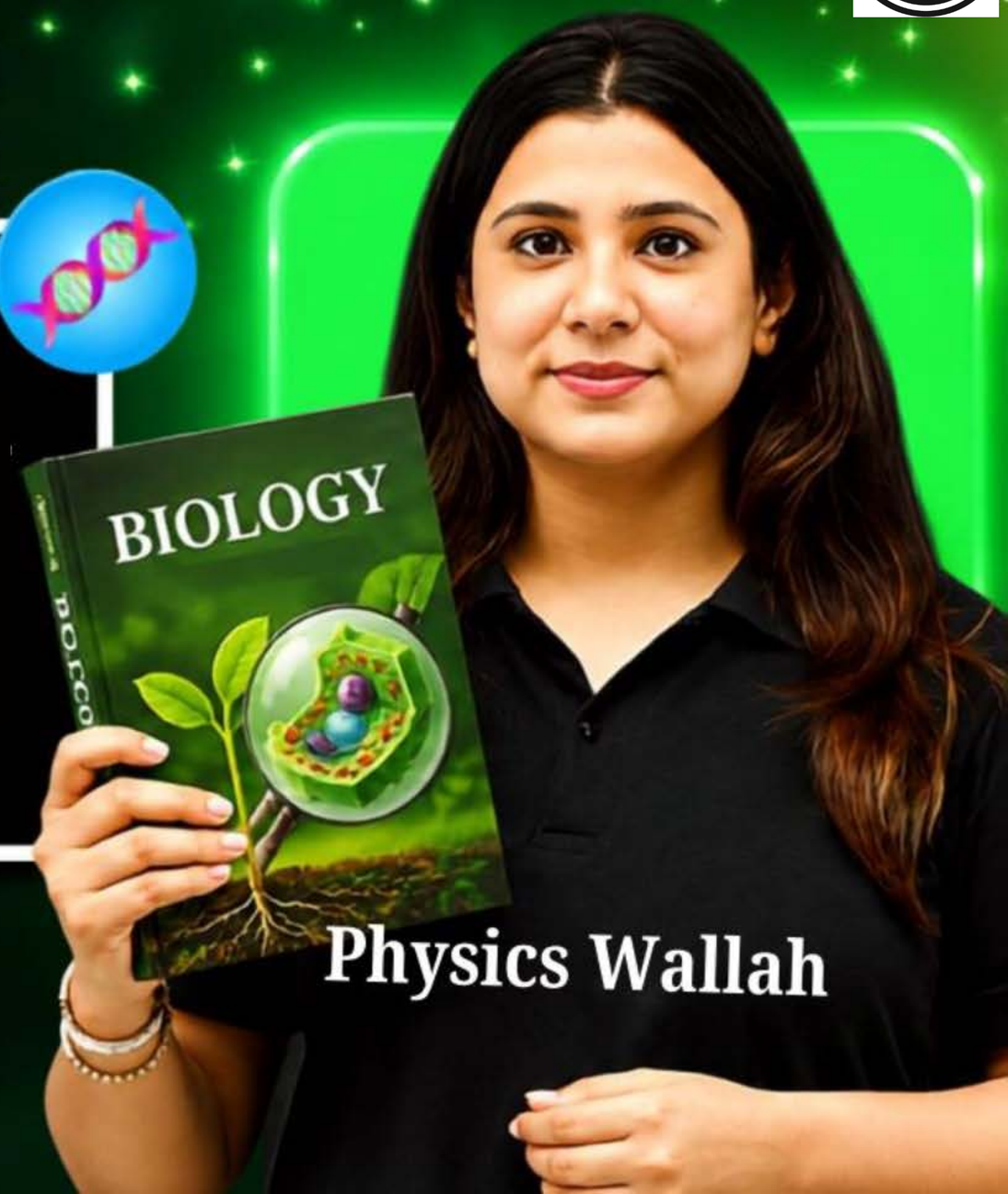
Victory 2027

CLASS 10TH ICSE

THE CIRCULATORY SYSTEM

LECTURE-04 BIOLOGY

By- Upasana Ma'am



Topics To Be Covered

1. White Blood Cells
2. Platelets
3.


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

Kaun Banega NINJAPATI ??



85++ लाना है

Q1

Red Blood Cells are produced in?

A. liver
B. Spleen
C. Red Bone Marrow ✓
D. Lymph Nodes

Q2

A decrease in RBCs result in.

A) Anaemia ✓

B) Diabetes

C) Leukemia

D) Thrombocytosis

Q3

Which metal ion helps in O_2 transport in Haemoglobin?

A) Ca

B) Fe ✓

C) Na

D) K

Q4.

Which organ is known as Graveyard of RBCs?

A. Heart
B. Spleen ✓
C. Kidney
D. Pancreas

Q5.

Mature RBCs lack ____

A) Nucleus

B) Cell Membrane

C) Haemoglobin

D) All of the above


CONGRATULATIONS

FORMATION OF ANTIBODIES

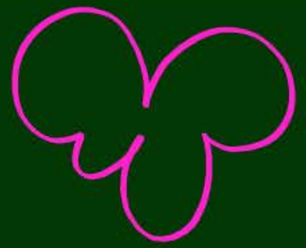
- The WBCs (specially the lymphocytes) produce antibodies which kill or neutralise the germs, or the poisons from them. Introducing weakened germs or germ substances (vaccines) during vaccination stimulates formation of particular antibodies which, at a later period, would destroy the particular disease-causing germs if they enter into the body.



Types of WBCs

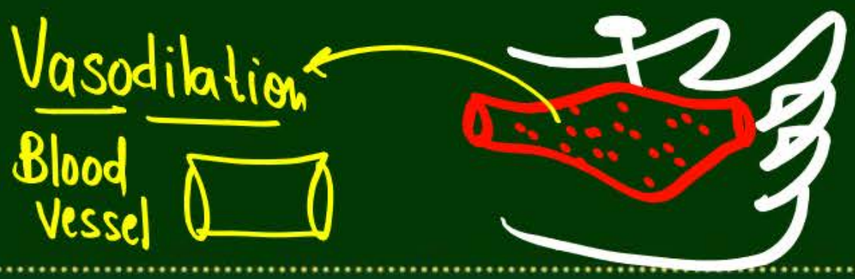
MLA → Agranulocytes
↓ ↳ Lymphocytes
Monocytes

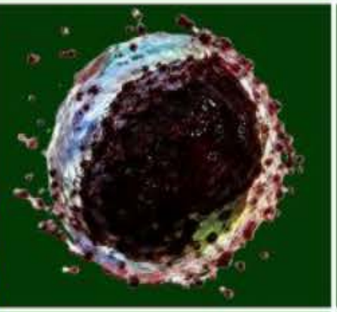
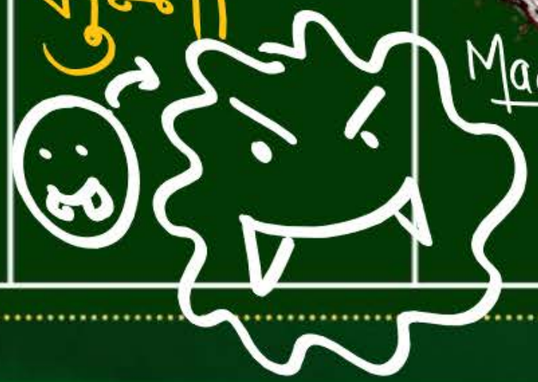
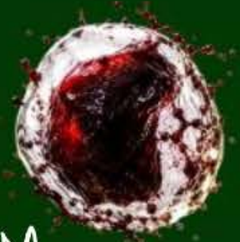
Two major categories of WBCs	Five cell types (Normal range)	Appearence	Distinguishing Features	Functions	Site of Production
A. Granular • Cytoplasm contains granules • Nucleus usually constricted into lobes	1. Neutrophils (55-70%) 1st line of (Defense)		• Nucleus with 3-4 lobes • Granular cytoplasm • Stain with neutral dyes	• Engulf bacteria • (Phagocytosis) 	• Bone marrow

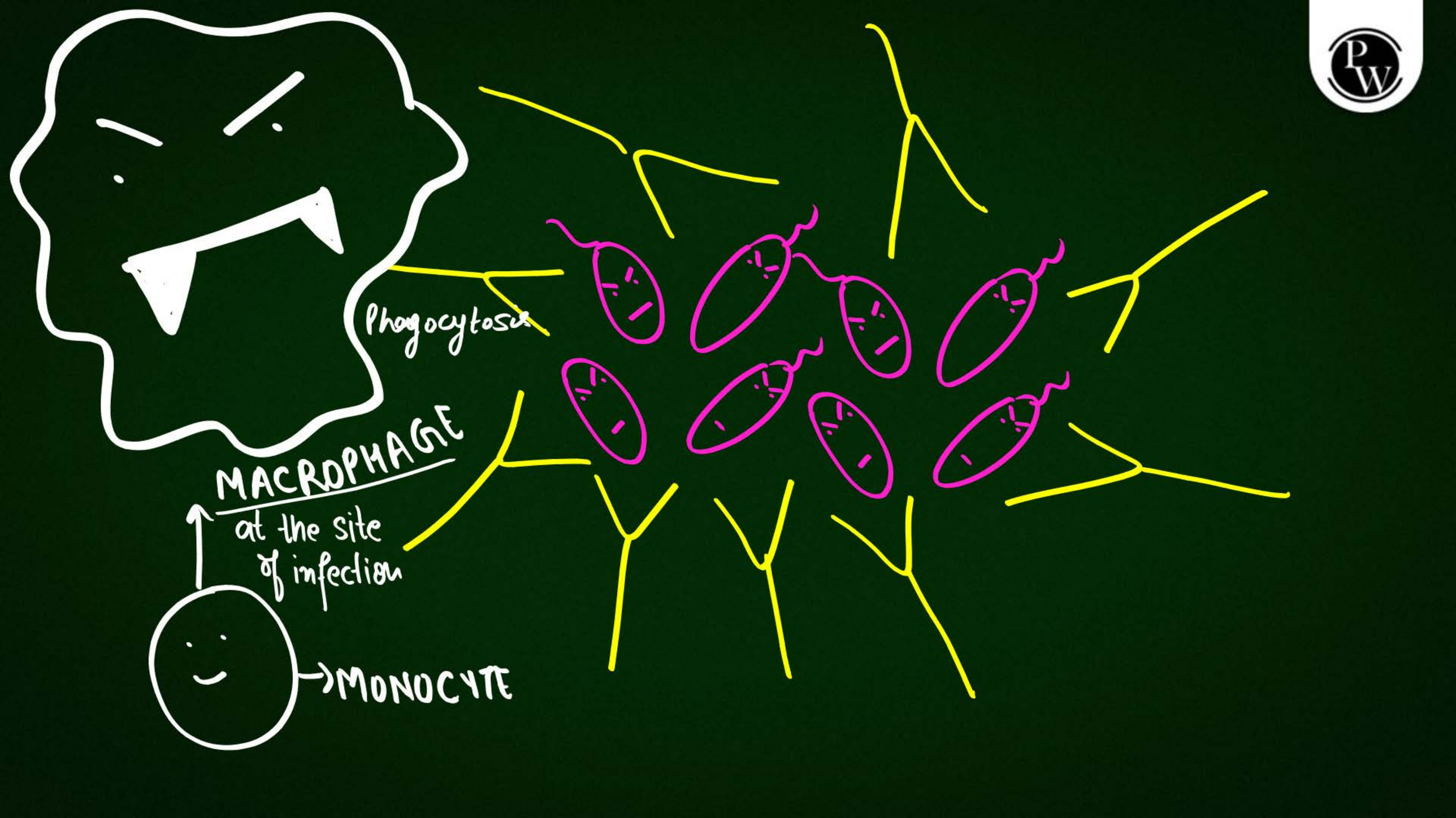


Two major categories of WBCs	Five cell types (Normal range)	Appearence	Distinguishing Features	Functions	Site of Production
	2. Eosinophils (1-3%) [count increases in allergies] (Acidophils)		• Nucleus with 2 lobes • Large cytoplasmic granules • Stain dark red with eosin (acid dye)	phagocytosis • Engulf bacteria • Secrete antitoxins • Associated with allergy	• Bone marrow

Two major categories of WBCs	Five cell types (Normal range)	Appearence	Distinguishing Features	Functions	Site of Production
	3. Basophils (0.5-1%) → Bilobed		• Nucleus large and indistinctly lobed • Granules stain with basic dyes (e.g., methylene blue)	• Release chemicals (histamine) for inflammation which dilate blood vessels Serotonin (Vasoconstriction)	• Bone marrow



Two major categories of WBCs	Five cell types (Normal range)	Appearence	Distinguishing Features	Functions	Site of Production
B. Non-granular • Cytoplasm without granules • A single large nucleus	4. Lymphocytes (20-35%) B cells → [Blood] T cells → [Tissue]		• Smallest of WBCs • Single large nucleus	• Produce antibodies Aey Chup Rch	• Bone marrow and Lymph glands (spleen, tonsils, etc.)
	5. Monocytes (3-8%) मुन्ना 	 Macrophage Large eating	• Large of WBCs • Nucleus large, • kidney-shaped • At the site of infection, transform into macrophage	• Ingest germs	• Bone marrow

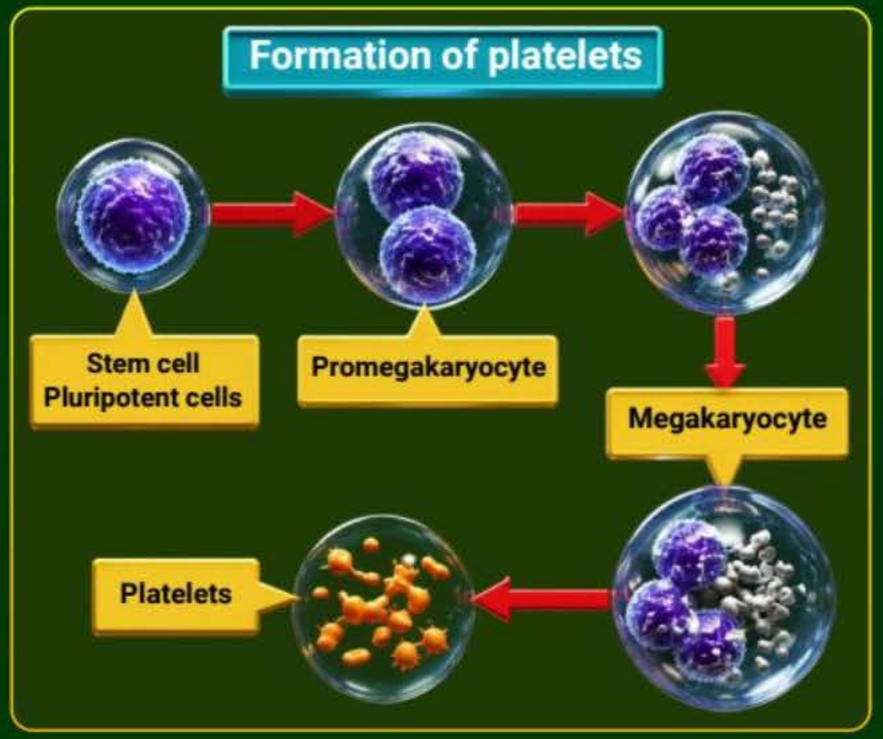

Phagocytosis
MACROPHAGE
at the site
of infection
MONOCYTE

TRICK

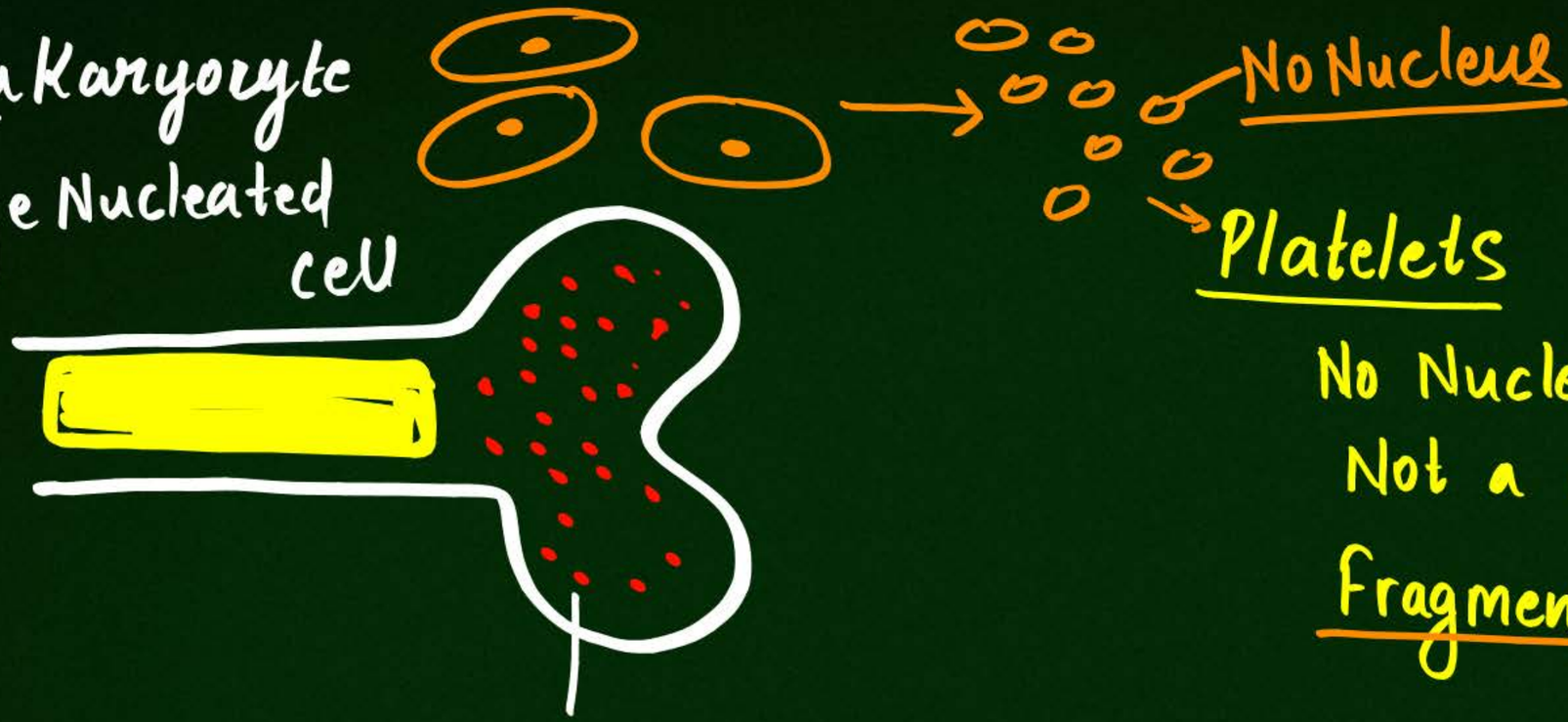
Never	Neutrophils 60%.
Let	Lymphocytes 30%.
Me	Monocytes 3% – 6%.
Eat	Eosinophils. 3%.
Bananas	Basophils 0.5-1%.

BLOOD PLATELETS (THROMBOCYTES)

- Blood platelets are minute oval or round structures, non-nucleated, floating in the blood. These are about 200,000 to 400,000 per cu. mm. of blood in an adult.
- The platelets are derived from giant cells called megakaryocytes in the red bone marrow.
- These are budded off from the megakaryocytes
- Their life span is 3 to 5 days and are destroyed mainly in the spleen (a lymphatic organ located in the blood.



MegaKaryocyte
large Nucleated cell



No Nucleus

Platelets

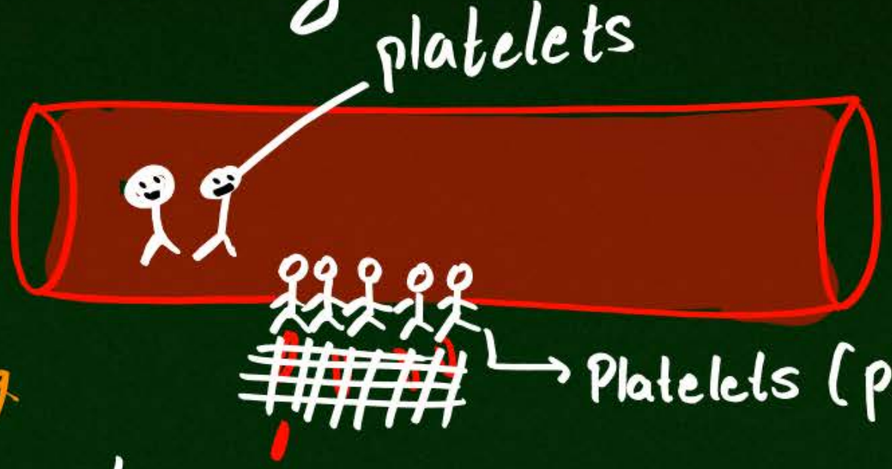
No Nucleus
Not a complete cell
Fragments of large MegaKaryocyte

Red Bone Marrow
→ RBC, WBC, Platelets formed here

Blood Clotting



Vit K in liver
Synthesise
enzymes for Clotting



→ Platelets (plug formation)

Thrombokinase/
Thromboplastin

Thrombokinase

Prothrombin → **Thrombin**

Ca^{2+}

Inactive





Mesh like structure.



Fibrinogen → **Fibrin**

Ca^{2+}

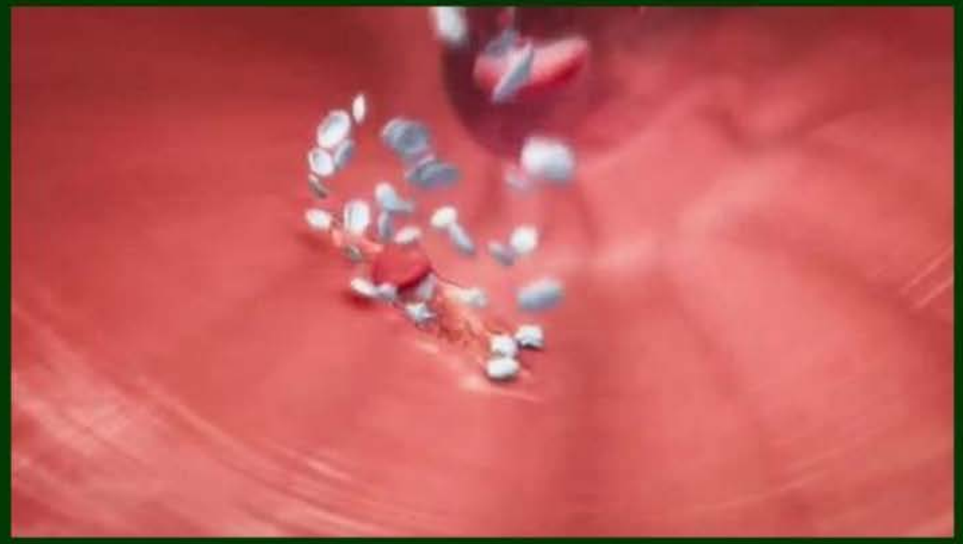
(Soluble, Inactive)

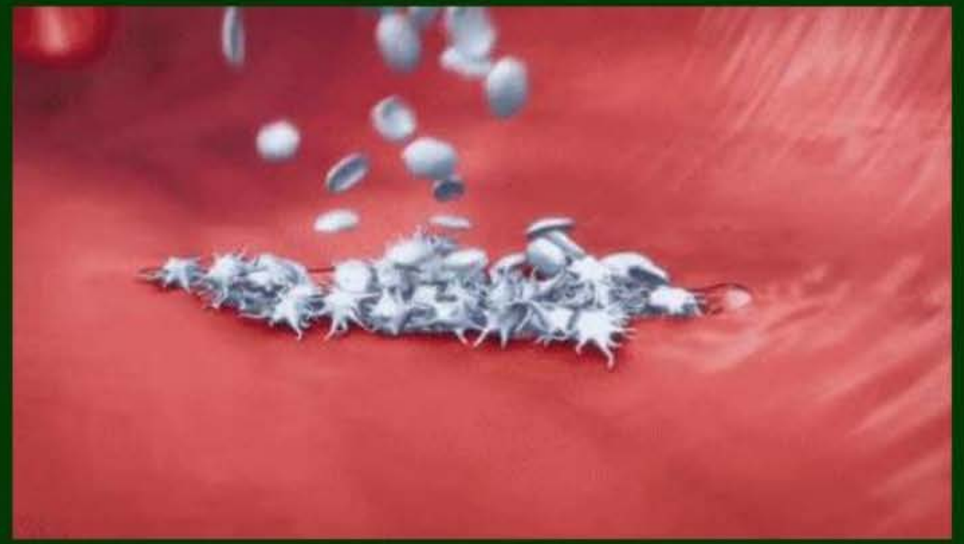
(Insoluble, Active)

FUNCTIONS OF PLATELETS CLOTTING OF BLOOD (COAGULATION)

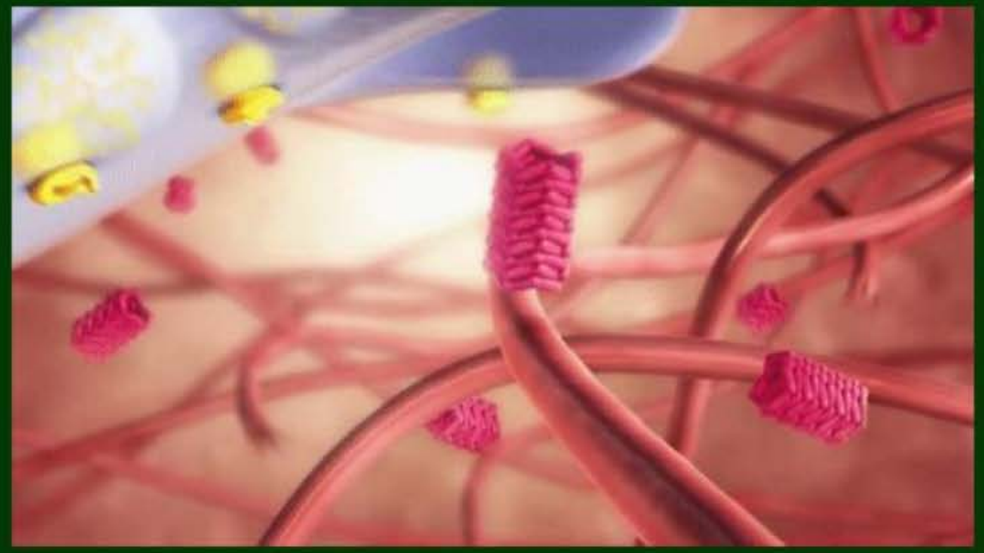
- Clotting (also called coagulation) occurs in a series of steps as follows:

1. The injured tissue cells and the platelets which disintegrate at the site of the wound release a substance thrombokinase (also called thromboplastin). (More recently it is called "Factor X" or Stuart factor).



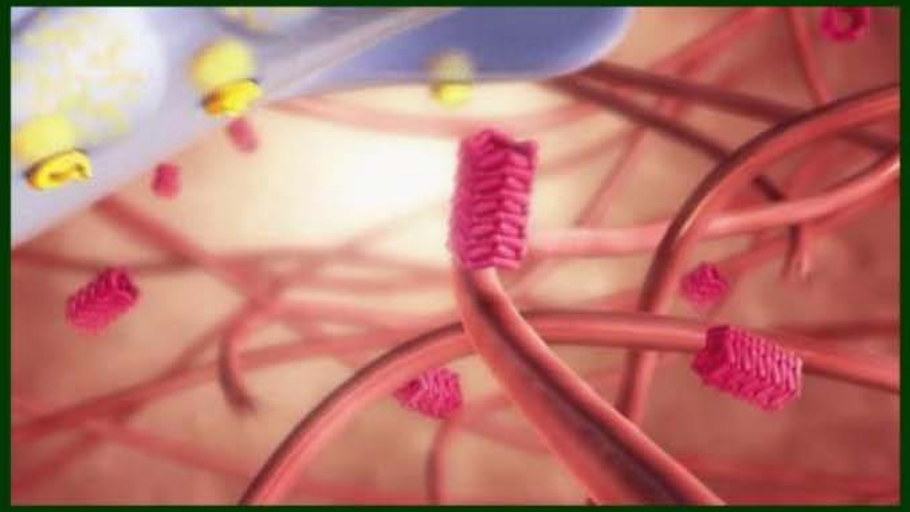


1. The thrombokinase acts as an enzyme and with the help of the calcium ions present in the plasma, it converts a substance **prothrombin** (inactive) of the plasma, into **thrombin** (active). **Vitamin K**, a fat-soluble vitamin, is essential for the production of **prothrombin**.



1. Thrombin in the presence of calcium ions, reacts with the soluble fibrinogen of the plasma to convert it into insoluble fibrin.

BLOOD CLOT [THROMBUS]





CLOTTING OF BLOOD

- Cascade process → 4 minute

- Step 1: Thrombocytes (Platelets) + Air → Thromboplastin
- Step 2: Thromboplastin + Ca + Prothrombin → Thrombin
- Step 3: Thrombin + fibrinogen → Fibrin
- Step 4: Fibrin (network) + Blood cells → Blood clotting → THROMBUS

I love all my NINJAS
Not all could connect through VC
I feel bad for
those who could
not join.

Though I enjoy VC
I hope you understand

Thank You

Yours Uppu Mummy~